

Nitte (Deemed to be University)

(Established under Sec.(3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CST101 – MECHANICS OF DEFORMABLE BODIES

(For CST)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. With suitable notations derive the equations of equilibrium in cylindrical coordinate system. | 20 | L3 | 1 | 1,2 |
| 2. At a point in a given material, the three-dimensional state of stress is given by $\sigma_x = \sigma_y = \sigma_z = 10 \text{ N/mm}^2$, $\tau_{xy} = 20 \text{ N/mm}^2$, and $\tau_{yz} = \tau_{zx} = 10 \text{ N/mm}^2$. Compute the principal planes if the corresponding principal stresses are $\sigma_1 = 37.3 \text{ N/mm}^2$, $\sigma_2 = -10 \text{ N/mm}^2$, $\sigma_3 = 2.7 \text{ N/mm}^2$ | 20 | L3 | 1 | 1,2 |

Unit – II

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|---|----|----|-----|-----|
| 3. The components of strain at a point in a body are as follows:
$\epsilon_x = 0.1$, $\epsilon_y = -0.05$, $\epsilon_z = 0.05$, $\gamma_{xy} = 0.3$, $\gamma_{yz} = 0.1$, $\gamma_{zx} = -0.08$. Determine the principal strains and the principal directions. | 20 | L3 | 1,2 | 1,2 |
| 4. The displacement components in a strained body are as follows:
$u = 0.01xy + 0.02y^2$,
$v = 0.02x^2 + 0.01z^3y$,
$w = 0.01xy^2 + 0.05z^2$.
Determine the strain matrix at the point P(3, 2, -5). | 20 | L3 | 1,2 | 1,3 |

Unit – III

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|---|----|----|---|-------|
| 5. Given the stress function: $\phi = -(F/d^3) xz^2 (3d - 2z)$. Determine the stress components and sketch their variations in a region included in $z = 0$, $z = d$, $x = 0$, on the side x positive. | 20 | L4 | 3 | 1,2 |
| 6. An infinite plate with a circular hole of radius a is loaded by tension σ_∞ along the X -axis. Using suitable notations, derive the stress variation around the hole and identify where the maximum stress occurs. | 20 | L4 | 3 | 1,2,4 |

Unit – IV

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| 7. A thin-walled steel section shown in figure is subjected to a twisting moment T . Calculate the shear stresses in the walls and the angle of twist per unit length of the box. |
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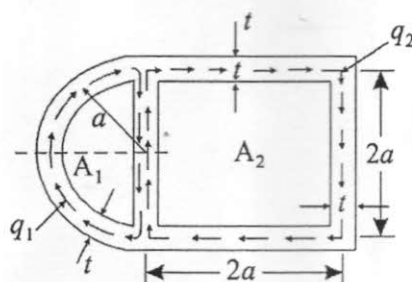
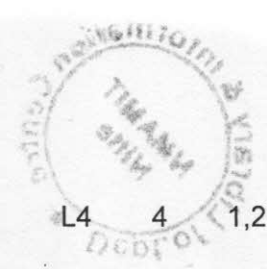


Fig. 7a

20	L4	4	1,2
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8. By considering the equations of equilibrium, find the general solution for compatibility for a bar subjected to torque.

20 L4 4 1,2

Unit – V

9. Compare and differentiate the significance, assumptions, and applicability of the following failure theories
(i) Maximum Elastic Energy Theory,
(ii) Octahedral Shearing Stress Theory,
(iii) Maximum Elastic Strain Theory,
(iv) Maximum Shearing Stress Theory, and
(v) Maximum Principal Stress Theory for predicting yielding in ductile and brittle materials under complex stress states.

20 L3 5 1,2

10. Analyze how different failure theories influence the design of a circular shaft subjected to combined bending moment M and torque T . Using a specified factor of safety N , determine the required diameter d of the shaft according to each theory, and discuss how the predicted diameters reflect the comparative conservatism of the theories.

20 L4 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
